Advanced C# Tasks - Day 1: Interface, Property

Task 1:

Goal: Create a PhoneBook class with indexers for storing contact information.

Step 1.1: Define a PhoneBook class with an indexer that allows both
phone number (integer) and contact name (string) as keys.
- Hint: Ensure your indexer handles both types of keys differently.
Do not simply use `object` as the type for both keys. Try something more creative and type-safe. How can you make sure to handle a string and an integer separately without using inheritance?
- You must use two different indexer signatures in your class.
Do not copy-paste a single line of code.

Step 1.2: In addition to adding contacts, implement validation logic
for invalid phone numbers or names. For example, phone numbers should only be integers greater than zero. How can you incorporate data validation within an indexer to ensure that only valid information is added to the PhoneBook?

Task 2:

Goal: Create a Person class with various types of properties, including automatic, read-only, and write-only properties.

Step 2.1: The Person class should have:
- A read-only property that combines FirstName and LastName into a single Full Name.
- An automatic property for Age.
- A write-only property for Password.

Task 3:

Goal: Implement a system that uses interfaces and demonstrates inheritance through concrete classes.

Step 3.1: Define an interface `IShape` with the following methods:
- `CalculateArea()`: Returns the area of the shape as a `double`.
- `CalculatePerimeter()`: Returns the perimeter of the shape as a `double`.

Step 3.2: Define another interface `IColor` with the following method:
- `GetColor()`: Returns the color of the shape as a `string`.

Step 3.3: Implement the following classes that implement the above interfaces:
- `Rectangle`: Properties should include `Width` and `Height`.
- `Square`: Properties should include `SideLength`.
- `Circle`: Properties should include `Radius`.